

Sahayak: from the application to the working prototype

The application (Inclusive Innovation for Bharat, Roshan Raj) described the idea and a clickable demo. The prototype submitted on 8 October 2026 is working software: a node, a phone web app and an operator console, with 360 automated tests and a benchmark behind every number. Building it changed some choices. Each change below was made because a measurement said so.

The application said	The prototype does	Why
A compressed 14B-class model, offline on the Crucible runtime, answers questions	51 named signals, a pattern matcher and a small classifier decide every verdict in about 2 ms. Vetted Hindi and English templates explain it. An optional local model (tested: qwen2.5:3b on llama.cpp) can add a sentence through the safety gate; it is off by default. The model interface is OpenAI-style, so Crucible or any local server can plug in	A verdict a juror can check, the same every time, on a cheap CPU. On the 4-core test laptop the 3B model took 11–21 s per answer and its sentences were weaker than the templates. Crucible has not been run with Sahayak yet
A fine-tuned FinTech-advisor model answers money questions	Scheme answers come from rules written as data, each traced to its official page and date. No model writes an eligibility answer or an amount	Eligibility and rupee amounts must never be invented. 2,400 of 2,400 decisions match an independent re-derivation. The Lenn Chartered fine-tune is past experience, not part of Sahayak
Every answer passes an evaluation gate	Every model sentence is checked before anyone sees it: dangerous advice, phone numbers or links not in the message, invented amounts, contradicting the verdict, denying a number or link that is there, wrong script	The same idea, built into the product and covered by tests. The benchmarks use the same rigour: frozen test split, 95% bootstrap intervals, McNemar's test, leakage check
Under 800 ms	2 ms per verdict; 0.65 s from releasing the mic to the spoken reply (2.44 s when the sentence is synthesised live)	Measured on a 4-core laptop CPU with no GPU
0 bytes outbound	An audit hook counts every connection and DNS lookup Sahayak makes to the internet, live on the node page: 0. Firewall scripts block the rest of the machine	Measured, not asserted. The node page also shows the machine's own internet connections, honestly
Voice-first, vernacular	Offline speech in and out in Hindi and English; Hinglish text; amounts read as words; "I heard ..., is that right?" before an answer is used	Two languages done properly first. Hindi word error 15.4% on Google's FLEURS test set. More languages are a content pack
~90% query resolution; 70–90% lower cost per user	Not claimed for Sahayak	These were not measured for Sahayak
Health and learning on the same platform	Not built. Finance only: scam checks and benefits	The flagship first
An interactive phone-style demo	Working software you can run, plus a jury kit of test cards with an answer key produced by Sahayak itself; after one visit the phone checks messages and runs the benefits interview by itself, offline, with answers identical to the node's on every benchmark case, and a stand-alone build runs in any phone browser	The prototype stage asked for a running product, and scams arrive at home, not at the counter

Added beyond the application: UPI QR checks ("a QR never brings money in"), screenshots read offline, phone-call descriptions, a 1930 complaint draft, an operator console with an "Ask the agent" queue and impact counters with no personal data, Ed25519-signed content packs, and a red team of 52 disguised scams (46 caught on the first run against 33 for a keyword blacklist; all 52 after the fixes it found).

Unchanged: the plan for a 6-month, ~20-village pilot through CSC and bank-correspondent points (budget estimate ₹35–40 lakh), and the partners we are asking for. The node's counters are how that pilot would report results.

Team: the application lists Roshan Raj; the prototype was built by Roshan Raj and Ishmiit Singh.